Project 2: Agent-Based Simulation of Ophiocordyceps unilateralis infection in a Carpenter Ant Colony

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"All these spores mean they've been here for a while."
[1]

1 Introduction

Ophiocordyceps unilateralis is a parasitic fungus that targets carpenter ants. When a foraging ant picks up fungal spores from the forest floor, the fungus hijacks the ant's nervous system over the course of several weeks. The ant eventually climbs to a fixed height above the ground, bites into a leaf vein in a rigid "death grip," and dies. The fungus then erupts from the ant's head and rains new spores downward onto the foraging trails below, ready to infect the next wave of ants. [2]

This system is a positive feedback loop as more infected ants mean more spores. More spores mean more infections. Whether this loop runs away and kills the colony, or burns itself out, depends on a race between how fast the infection spreads and how fast the colony reproduces.

Rather than writing a single equation for the average number of infected ants (like the classical epidemic model), we track every individual ant separately. Each ant is a software object with its own state (egg, larva, adult, infected, zombie, sporulating, or dead) and its own age. At each simulated day, every ant makes its own probabilistic decisions: did this forager pick up a spore today? Did this infected ant succumb overnight? This style of simulation is called agent-based because each ant acts independently, and the colony-level outcome emerges from the combined behavior of all of them. It is also stochastic because the outcomes are random: two runs of the simulation with identical starting conditions but different random seeds will produce different histories.

Our goal is to understand what determines whether a colony survives a cordyceps outbreak. We examine the effect of three quantities: the infection rate p (how easily spores infect a forager), the initial colony size N_0, and the immune resistance (how long an infected ant can fight off the parasite before succumbing). We also compare the stochastic agent-based model against a deterministic ODE counterpart. Figure 1 shows a normal colony before any infection is introduced.

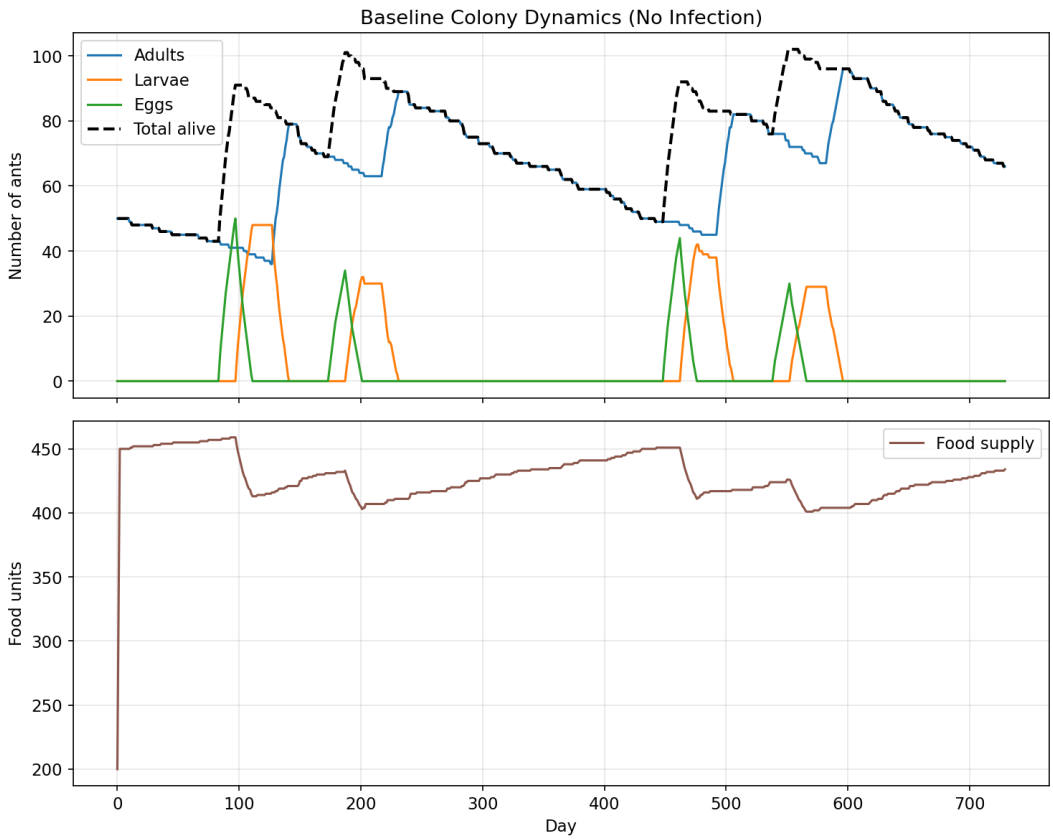

Figure 1: Normal colony dynamics with no infection ($N_0 = 50$ adults, $F_0 = 200$ food units, run for 730 days)

Adults (blue) decline gradually between the yearly egg-laying pulses. Larvae (orange) and eggs (green) appear at those pulses and then mature according to their fixed developmental schedules, while food in the lower panel remains comfortably above zero throughout.

2 Governing Algorithm

This simulation has no differential equations in its overall algorithm as every event in the model is defined to take a whole number of days time advances in fixed one-day steps. Models of this kind are governed by an algorithm rather than by equations of motion.

2.1 The unknowns

At the start of day n, the quantities being tracked are:

- The state and age of every ant in the simulation. The state s_i of ant i is one of:

$$s_i \in \{\texttt{egg}, \texttt{larva}, \texttt{adult}, \texttt{infected}, \texttt{zombie}, \texttt{sporulating}, \texttt{dead}\}.$$

  Each ant also carries d_i, the number of days it has spent in its current state.

- The **food supply** $F_n \geq 0$, measured in food units (one unit feeds one ant for one day).

- The **environmental spore concentration** $c_n \geq 0$, a number representing the density of viable spores in the foraging area. It rises when sporulating corpses are present and decays otherwise.

Figure 2 summarizes all possible ant states and the transitions between them.

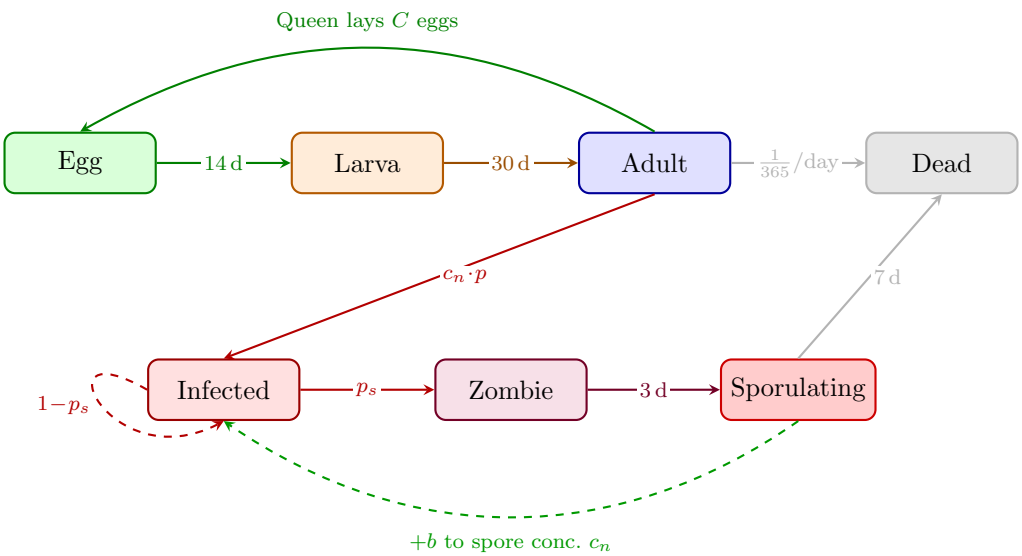

Figure 2: Individual ant state diagram.

The top row shows the normal lifecycle of an ant, from egg to adult and eventually to natural death. The bottom row shows the infection pathway. Infected ants continue to forage and eat, so they remain indistinguishable from healthy ants within the colony. The dashed green arrow represents the positive feedback loop where sporulating corpses increase c_n, which in turn drives new infections.

2.2 Fixed parameters

The following quantities are inputs to the model. They are not computed by the simulation but are chosen before it begins.

Symbol	Value	Meaning
δ	1/365	Daily natural-death probability for any non-egg ant
a	15	Foraging constant (sets typical yield per forager)
k_{egg}	8	Egg-laying scaling constant
b	0.005	Spores added per sporulating ant per day
k	0.30	Daily fractional decay of spore concentration
T_{egg}	14 days	Incubation time from egg to larva
T_{larva}	30 days	Development time from larva to adult
T_{zombie}	3 days	Time as zombie before sporulating
T_{spore}	7 days	Duration of sporulation before the corpse is spent
p	(varied)	Infection probability per unit of spore concentration
p_s	0.05	Daily probability that an infected ant succumbs
c_0	0.05	Spore concentration set on the introduction day

The top block is fixed across all runs. The bottom block is scenario-dependent and are varied for our parametric studies in Section 6.

2.3 The daily update: seven steps

On each day, the following steps are applied in order to advance from day n to day $n+1$.

Step 1: Lifecycle transitions. Any egg that has been an egg for at least $T_{\text{egg}} = 14$ consecutive days becomes a larva. Any larva that has been a larva for at least $T_{\text{larva}} = 30$ consecutive days becomes an adult.

Step 2: Foraging. Each adult ant and each infected adult goes out and brings back food. The amount brought back by forager i is a random draw from an exponential distribution:

$$F_i \;\sim\; \text{Exp}\Big(\text{mean} = a\, N_n^{-1/3}\Big).$$

The mean per-forager yield $a\, N_n^{-1/3}$ decreases as the colony grows, reflecting the fact that more foragers deplete the surrounding area faster, a form of diminishing returns. All food is pooled and then capped:

$$F \;\leftarrow\; \min\Big(F_n + \textstyle\sum_{\text{foragers}} F_i,\; F_{\max}\Big).$$

Step 3: Eating. Every larva, adult, and infected adult consumes one food unit. (Eggs do not eat whereas zombies and sporulating ants have already stopped caring about food.) If there is not enough food for everyone, a randomly chosen subset of the hungry ants is fed and the rest die of starvation immediately.

Step 4: Natural death. Every larva, adult, and infected adult dies from natural causes with probability $\delta = 1/365$, independently. This models background mortality from predation, aging,

and accident, calibrated so that the mean lifespan of an adult that avoids all other causes of death is one year.

Step 5: Egg-laying. The queen lays eggs twice a year: in spring (days 83–96 of the annual cycle) and in summer (days 173–186). The number of eggs laid each day within a window is:

$$C = \left\lfloor \min\left(k_{\text{egg}} \frac{F_n}{N_n} \div 14,\ \frac{100}{14} \right) \right\rfloor$$

Dividing by 14 spreads the seasonal clutch evenly across the window. The F_n/N_n ratio ties reproduction to colony health: a colony short on food or overburdened with mouths lays fewer eggs. The cap of 100 total eggs per clutch prevents runaway reproduction.

Step 6: Infection and spore dynamics. This step is only active after spores have been introduced. It has four sub-steps:

(a) New infections. Each healthy adult that is currently foraging is exposed to the spore-laden environment. It becomes infected with probability

$$P(\text{infected}) = \min(c_n \cdot p,\ 1).$$

The probability is proportional to both the ambient spore level c_n and the species-level susceptibility p. The minimum with 1 prevents the expression from exceeding a valid probability at high concentrations.

(b) Immune mini-game. Each infected ant independently "rolls" each day. With probability p_s it *succumbs* and transitions to zombie; with probability $1-p_s$ it holds on for another day. There is no recovery. Biologically, this models the immune system slowing but not stopping fungal growth. The expected number of days from infection to succumbing is $1/p_s = 20$ days (at our default $p_s = 0.05$).

(c) Zombie and sporulation timers. Any zombie that has been a zombie for $T_{\text{zombie}} = 3$ days transitions to sporulating. Any sporulating ant that has been sporulating for $T_{\text{spore}} = 7$ days transitions to dead (fully spent; no more spores).

(d) Spore concentration update. Let S_n = number of currently sporulating ants. The concentration for the next day is:

$$c_{n+1} = c_n(1-k) + b\,S_n.$$

The first term models spore decay by a factor $k = 0.30$ per day. The second term adds $b = 0.005$ units of fresh spore per sporulating corpse. If $S_n = 0$ for all n, this simplifies to $c_n = c_0(1-k)^n$, giving a half-life of $\ln 2/\ln(1/0.70) \approx 1.9$ days meaning spores clear quickly without active sources.

Step 7: Age advance. Every ant increments its d_i (days in current state) by one. This is what triggers the lifecycle transitions in Step 1 on subsequent days.

The key behavioral differences between the infection-related states are:

State	Forages?	Eats?	Contributes to c_n?
Adult	Yes	Yes	No
Infected	Yes	Yes	No
Zombie	No	No	No
Sporulating	No	No	Yes
Dead	No	No	No

An *infected* ant behaves exactly like a healthy adult: it forages, eats, and spreads no spores itself. The colony has no way to detect or isolate it. Only when the ant succumbs and becomes a zombie does it withdraw from colony life.

3 Numerical Method

As the governing algorithm already advances time in discrete steps of exactly one day, there are no differential equations that need to be discretized. Here, the one-day timestep is not an approximation but rather built into the model by assumption. The algorithm is the numerical method.

The one exception is the spore concentration update $c_{n+1} = c_n + b\,S_n - k\,c_n$, which is forward Euler applied to the continuous ODE $dc/dt = b\,S(t) - k\,c(t)$ with $\Delta t = 1$ day. Since the spore half-life is about 1.9 days ($k = 0.30$), a 1-day timestep resolves the decay comfortably.

We then made the choice of the order in which the seven steps are applied within a single day. We apply foraging before eating, meaning a forager's haul contributes to the food pool on the same day the colony feeds from it. This is consistent with an interpretation where ants leave at dawn and return before feeding time.

3.1 Deterministic ODE model

To compare against the stochastic simulation, we replace every coin flip with its expected value. Instead of looping over N ants and rolling rand $< p$, we compute $p \cdot A$ new infections per step. Instead of giving each egg a fixed 14-day timer, we model eggs as leaving the egg stage continuously at a rate of E/T_{egg}. The resulting ODEs track eight state variables:

$$\frac{dE}{dt} = \text{laying_rate}(t) - \frac{E}{T_{\text{egg}}} \tag{1}$$

$$\frac{dL}{dt} = \frac{E}{T_{\text{egg}}} - \frac{L}{T_{\text{larva}}} \tag{2}$$

$$\frac{dA}{dt} = \frac{L}{T_{\text{larva}}} - \frac{A}{365} - c \cdot p \cdot A \tag{3}$$

$$\frac{dI}{dt} = c \cdot p \cdot A - p_s \cdot I - \frac{I}{365} \tag{4}$$

$$\frac{dZ}{dt} = p_s \cdot I - \frac{Z}{T_{\text{zombie}}} \tag{5}$$

$$\frac{dP}{dt} = \frac{Z}{T_{\text{zombie}}} - \frac{P}{T_{\text{spore}}} \tag{6}$$

$$\frac{dc}{dt} = b \cdot P - k \cdot c \tag{7}$$

$$\frac{dF}{dt} = (A + I) \cdot a \cdot N^{-1/3} - (L + A + I), \quad F \leq F_{\max} \tag{8}$$

where $N = E + L + A + I$ is the living colony size and the laying rate uses the same seasonal windows as the stochastic model.

Equations (1)–(3) describe the healthy colony where eggs are laid at a seasonal rate and hatch after T_{egg} days, larvae mature after T_{larva} days, and adults die naturally at rate $1/365$ day^{-1} while losing members to infection at rate $c \cdot p$.

Equations (4)–(6) describe the infection where infected ants transition to zombies at rate p_s (or die naturally), zombies burst into spore-releasing corpses after T_{zombie} days, and corpses decay after T_{spore} days.

Equation (7) governs spore concentration c where corpses emit spores at rate b while spores decay at rate k.

Equation (8) tracks the food stockpile where adults and infected ants forage at a rate scaling with $N^{-1/3}$, while all non-egg individuals consume food, capped at $F_{\max}$.

We advance the system with forward Euler at $\Delta t = 1$ day, matching the stochastic model's timestep. The key difference is that the ODE allows fractional ants (a colony of 0.3 adults can recover to 50) while the stochastic model enforces integer counts, creating a hard extinction threshold the ODE cannot capture.

4 Validation

4.1 Baseline colony with no infection

The fundamental check is to run the simulation with infection disabled and verify that colony dynamics match biological expectations. Figure 1 (Section 1) shows a 730-day no-infection run. The results are qualitatively correct:

- Adults decline slowly between seasons at a rate consistent with $\delta = 1/365$ per day.
- Two egg-laying pulses appear per year (near days 90 and 180), followed by larva peaks 14 days later and adult surges 30 days after that.
- Food never drops to zero, confirming the foraging model supports a colony of this size.
- The colony size after two years is approximately the same as the starting size, consistent with a roughly stable population.

4.2 Food balance at every step

On any day in which no ant starves, the food update is fully deterministic given the foraging amounts drawn:

$$F_{n+1} = \min\bigl(F_n + \Sigma_{\text{foraged}} - N_{\text{eating}},\ F_{\max}\bigr).$$

We verified this identity at every time step of several no-infection runs by recording the food before and after the step and comparing against the sum of drawn foraging values and the number of eating ants.

4.3 Stochastic variability across runs

Figure 3 shows 50 independent runs of the same scenario ($p = 0.15$, $c_0 = 0.05$, spores introduced on day 200, $N_0 = 50$). The spread illustrates the fundamental randomness of the model as some realizations see the outbreak fizzle early and others sustain elevated infection for months. This is not an error as it is how a stochastic model is supposed to behave. However, it means that a single run is not representative of the model's average behavior. All parametric studies in Section 6 therefore average over at least 20 independent trials per parameter value.

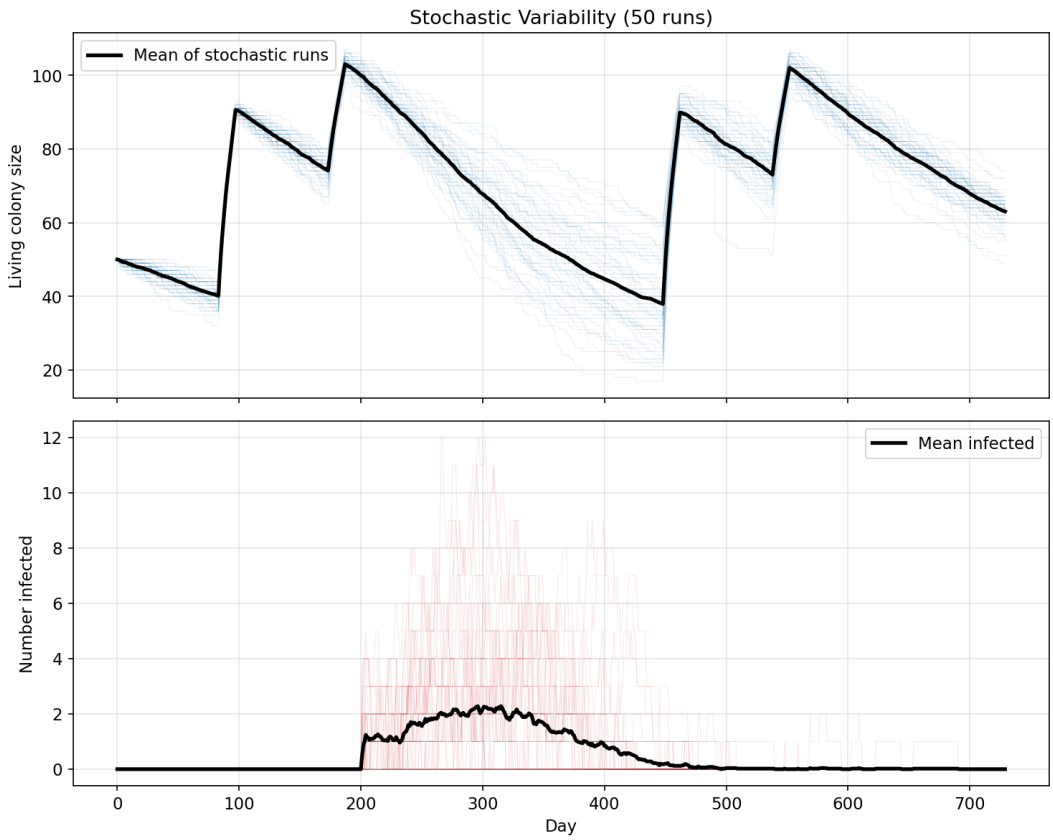

Figure 3: Living colony size across 50 independent runs at $p = 0.15$, $c_0 = 0.05$ (spores introduced on day 200)

5 Interesting Aspects of the Code

5.1 Behavior rules on the Ant object

Each ant's behavior is determined by three Boolean methods on the `Ant` class, encoding the rules from the table in Section 2.3:

```
def is_alive(self):
    # False for dead, sporulating and zombie
    # Sporulating ants still release spores, so they are and handled separately
    return self.state not in (AntState.DEAD,
                              AntState.SPORULATING,
                              AntState.ZOMBIE)

def consumes_food(self):
    # Larvae, healthy adults, and infected adults eat.
    # Zombies have stopped caring about food.
    return self.state in (AntState.LARVA,
                          AntState.ADULT,
                          AntState.INFECTED)
```

```python

def can_forage(self):
    # An infected ant still goes out to forage.
    return self.state in (AntState.ADULT, AntState.INFECTED)
```

The advantage of this design is that all five simulation phases (foraging, eating, etc.) interact with ants only through these predicates. The foraging phase, for instance, is simply `[a for a in self.ants if a.can_forage()]`. If we wanted to add a quarantine state in which an ant is kept away from foraging, we would only need to change `can_forage`, and all other phases would automatically respect the new rule.

5.2 The infection phase in full

The most complex part of the simulation is the daily infection phase, which implements all four sub-steps of Step 6 from Section 2.3:

```python
def _phase_infection(self):
    if not self.cordyceps_enabled:
        return

    # On the introduction day, initialize the concentration.
    if self.day == self.spore_intro_day:
        self.spore_concentration = self.c0

    # Nothing to do if there are no spores present.
    if self.spore_concentration <= 0:
        return

    # (a) New infections: each healthy foraging adult is
    #     exposed to the environment and may pick up spores.
    for ant in self._foraging_ants():
        if ant.state == AntState.ADULT:
            p_infect = min(self.spore_concentration
                           * self.infection_rate_p, 1.0)
            if self.rng.random() < p_infect:
                ant.transition_to(AntState.INFECTED)

    # (b) Immune mini-game: no recovery. Each infected ant
    #     independently rolls to succumb or hold on.
    for ant in self.ants:
        if ant.state == AntState.INFECTED:
            if self.rng.random() < self.p_succumb:
                ant.transition_to(AntState.ZOMBIE)

    # (c) Fixed-duration timers advance each state.
    for ant in self.ants:
        if (ant.state == AntState.ZOMBIE
                and ant.days_in_state >= Ant.ZOMBIE_DURATION):
            ant.transition_to(AntState.SPORULATING)

```

```
    for ant in self.ants:
        if (ant.state == AntState.SPORULATING
                and ant.days_in_state >= Ant.SPORULATION_DURATION):
            ant.transition_to(AntState.DEAD)

    # (d) Update spore concentration: decay + new spores
    #     from currently sporulating ants.
    n_sporulating = self._count_state(AntState.SPORULATING)
    self.spore_concentration = (
        self.spore_concentration * (1.0 - self.spore_decay_rate)
        + self.spore_birth_rate * n_sporulating
    )
    # Set to zero if rounding makes it slightly negative.
    self.spore_concentration = max(self.spore_concentration, 0.0)
```

6 Results and Discussion

6.1 Three qualitative infection regimes

Mild infection ($p = 0.08$): colony survives and recovers. Figure 4 shows a lightly contaminated environment. The outbreak flares briefly near day 200 (a few ants become infected, a few sporulating corpses appear, spore concentration ticks up) but then burns out. The feedback loop cannot sustain itself because at $p = 0.08$ the rate of new infections each day is smaller than the rate at which sporulating ants expire. Spores fall to near zero, and by day 300 the colony has returned to its normal seasonal rhythm. At the population level, the outbreak is barely visible.

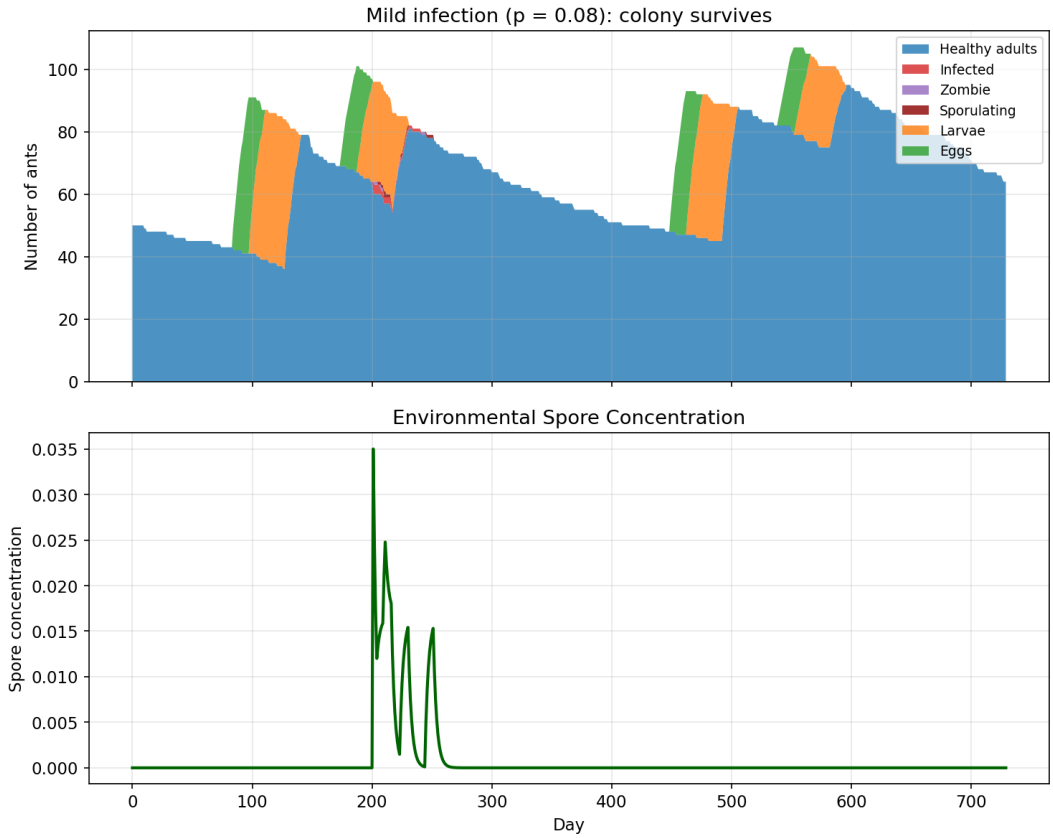

Figure 4: Mild infection ($p = 0.08$, $c_0 = 0.05$, spores introduced day 200)

Moderate infection ($p = 0.18$): colony weakened, persists for a year, then collapses. Figure 5 shows the intermediate case. After spore introduction on day 200, the infection neither fizzles nor explodes immediately. A persistent fraction of adults remain infected throughout, and the spore concentration oscillates at a low but nonzero level. Eventually the spores die out, but not before the adult population has been reduced enough that foraging output can no longer sustain the larvae. The colony then collapses from starvation rather than from infection directly.

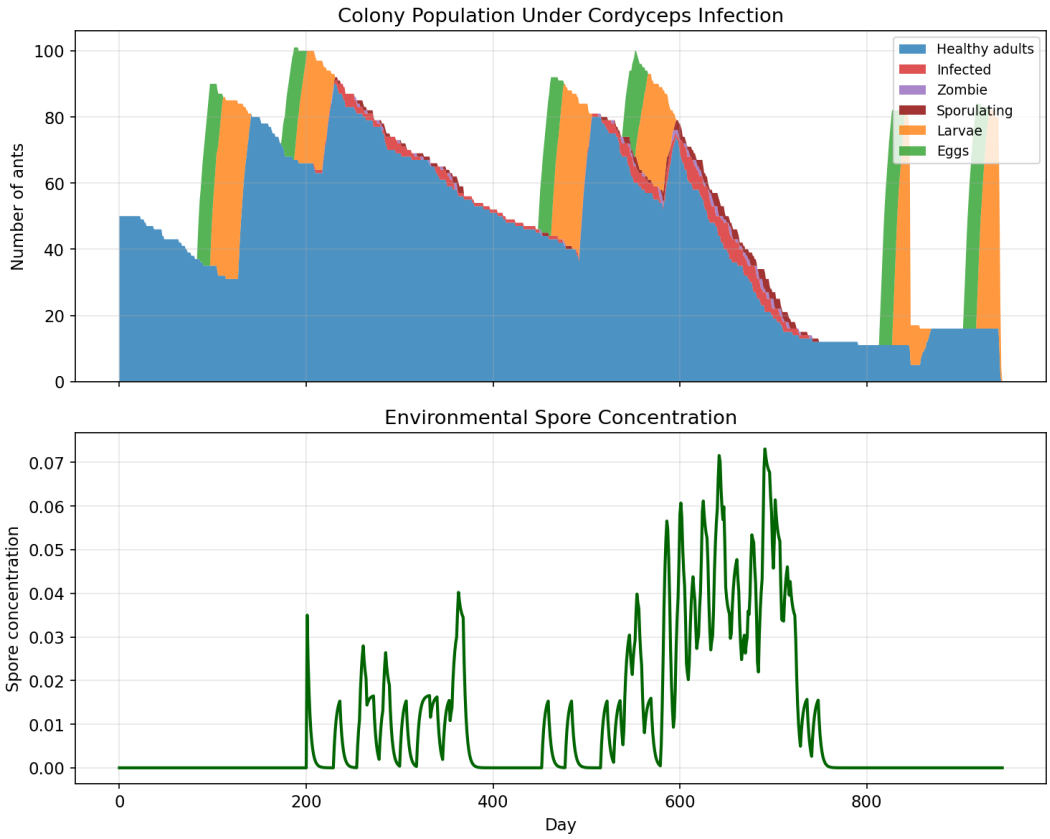

Figure 5: Moderate infection ($p = 0.18$, $c_0 = 0.05$, spores introduced day 150)

Severe infection ($p = 0.35$): colony collapses. Figure 6 shows the fatal case. After spore introduction on day 200, the feedback loop accelerates rapidly. Within a few months the sporulating population is large enough to drive c_n to high values, which in turn infects nearly all remaining healthy foragers. Without foragers, food supply collapses. Eggs and larvae starve. The queen cannot lay new eggs. Total extinction occurs by around day 400–500. The spore concentration shows a sharp peak and then a crash, because the colony itself was the fungus's food source, and once the colony is gone, the spores have nothing left to sustain them.

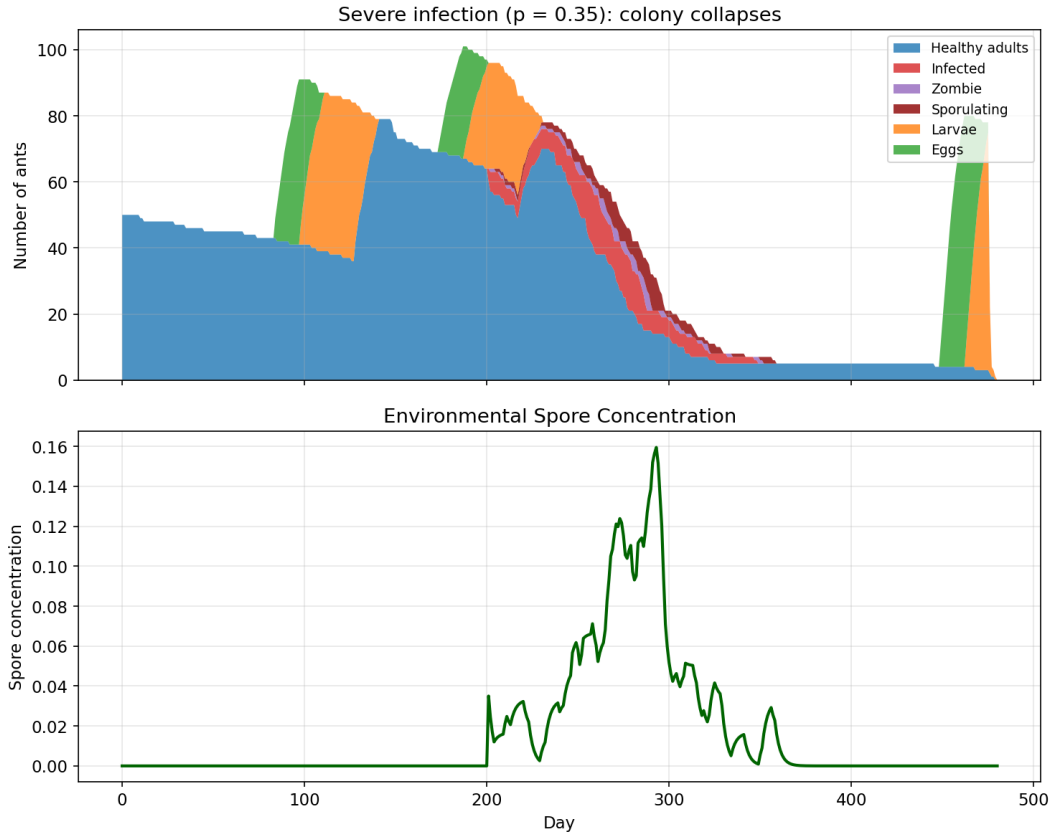

Figure 6: Severe infection ($p = 0.35$, $c_0 = 0.05$, spores introduced day 200)

6.2 Parametric study: does initial colony size help?

The natural intuition is that a larger colony should be more resilient, since it can lose some members without losing essential functions. Figure 7 tests this intuition and shows the opposite as, under the same infection rate ($p = 0.15$), a colony starting at $N_0 = 1000$ collapses faster and more reliably than one starting at $N_0 = 50$.

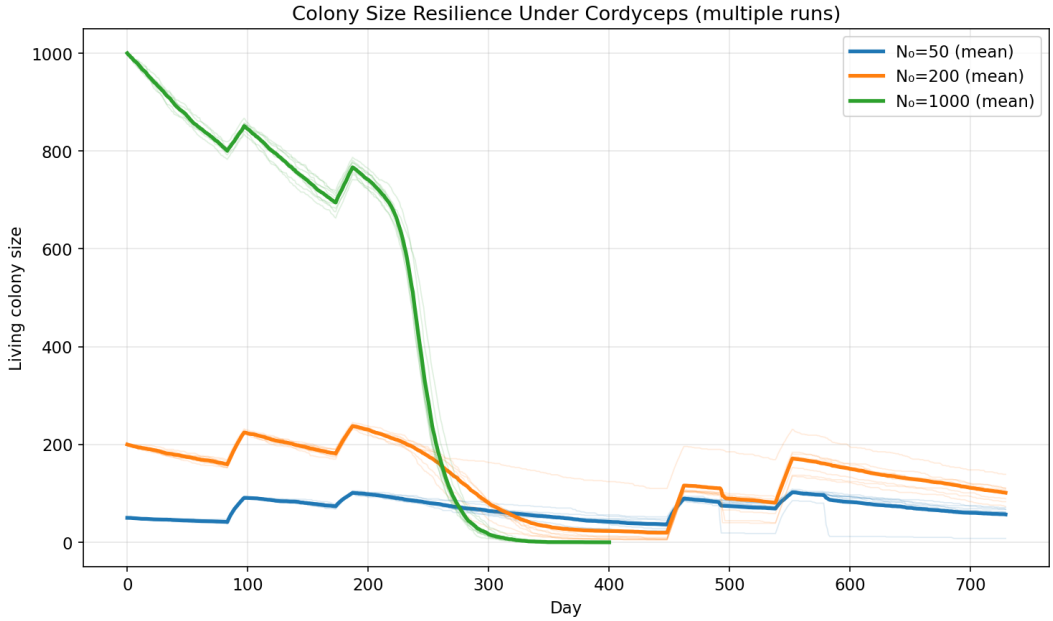

Figure 7: Living colony size vs. time for three initial colony sizes, all at $p = 0.15$

This is due to the $N^{-1/3}$ foraging model. A large colony sends out many foragers each day, exposing many ants to spores simultaneously. This produces a large initial infection wave that transitions into a large sporulating population, which drives c_n to high values before the colony can mount a demographic response. Meanwhile, each individual forager brings back less food at large N, so the food buffer that would support egg-laying during an outbreak is thinner. The small colony, by contrast, produces only a handful of sporulating ants at once. Spores never concentrate enough to sustain the loop, and the outbreak burns out.

6.3 Parametric study: how important is immune resistance?

Figure 8 sweeps the daily succumb probability p_s from 0.01 (mean 100 days to succumb; strong immunity) to 0.15 (mean 7 days; weak immunity), measuring colony survival probability at day 730 averaged over 20 trials per value.

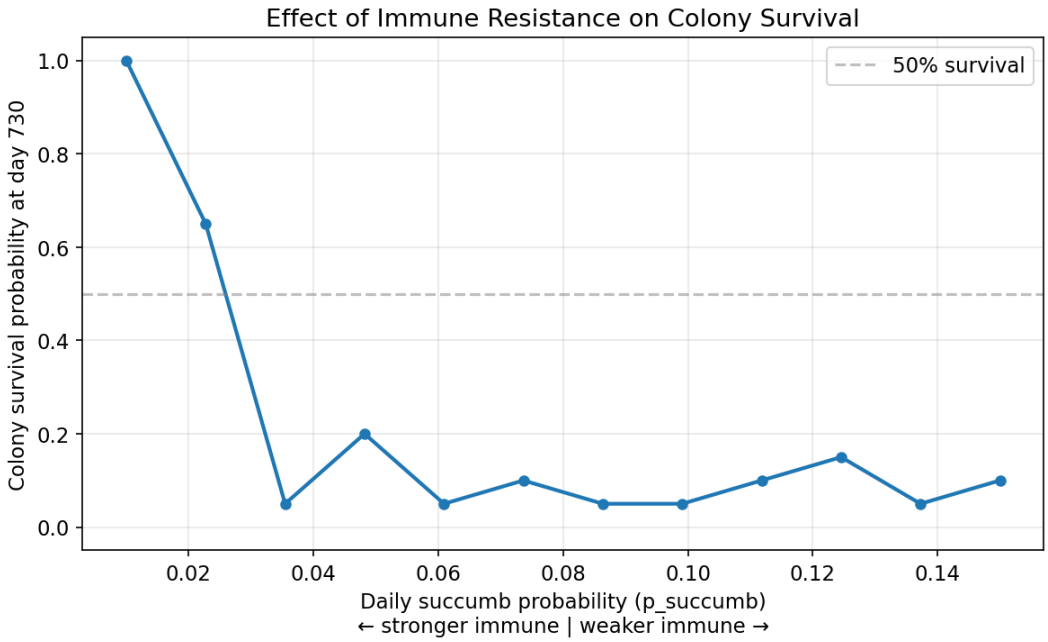

Figure 8: Colony survival probability at day 730 as a function of daily succumb probability p_s

The transition from near-certain survival to near-certain extinction happens in a narrow band around $p_s \approx 0.02$–0.04. At $p_s = 0.01$, infected ants survive about 100 days on average, long enough that natural death or starvation claims them before they ever become zombies, and the spore concentration never builds. At $p_s = 0.05$ (our baseline), the 20-day expected wait is short enough that most infected ants do succumb and sporulate. Above $p_s \approx 0.04$, survival probability is low but not zero: stochastic variation occasionally lets the outbreak fizzle before it gains momentum.

6.4 Parametric study: Threshold behavior over a range of infection rates

Infection prevalence (Figure 9) is the time-averaged fraction of the living colony that is infected, measured over the entire post-introduction period. It stays near zero for $p < 0.15$ and rises steeply above that. In an endemic disease model, the critical infectivity marks the boundary below which the disease dies out in the growing population. Our simulation reproduces this threshold behavior without computing R_0 explicitly.

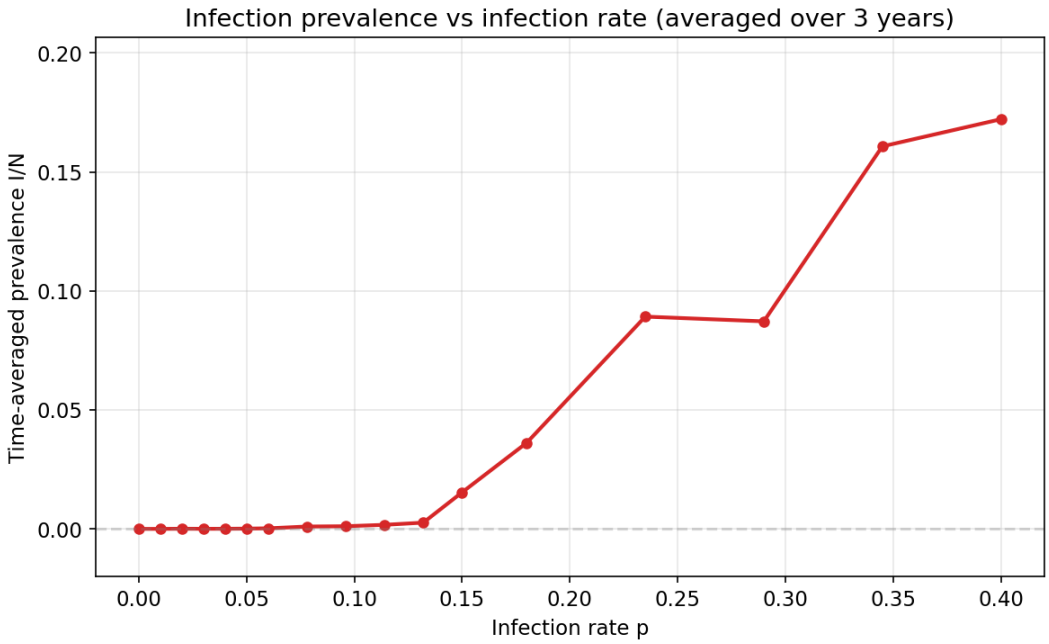

Figure 9: Time-averaged fraction of the colony that is infected vs. infection rate p

6.5 Stochastic vs. deterministic comparison

Figures 10 and 11 tests how the stochastic and deterministic simulations should agree for large populations, but diverge when the population size is small. This test was conducted by running 30 stochastic trials alongside the deterministic ODE system (1)–(8) at two colony sizes.

This comparison validates our stochastic model in two ways. First, the convergence at large N confirms that the agent-level rules are consistent with the ODE equations. Second, the divergence at small N demonstrates the need for an agent-based approach. For instance, for a colony of 50 ants, the ODE does not capture whether the colony dies out or persists.

At $N_0 = 50$ (Figure 10), the agreement is less apparent. The deterministic food curve sits at the cap of 500 (because the mean foraging rate exceeds consumption at this colony size), while the stochastic mean food is lower because of the variance in food amounts gathered from foraging. The infected and spore graphs show the right qualitative shape, but differ in timing and magnitude. This confirms that the stochastic effects by small colonies are significant and cannot be averaged away.

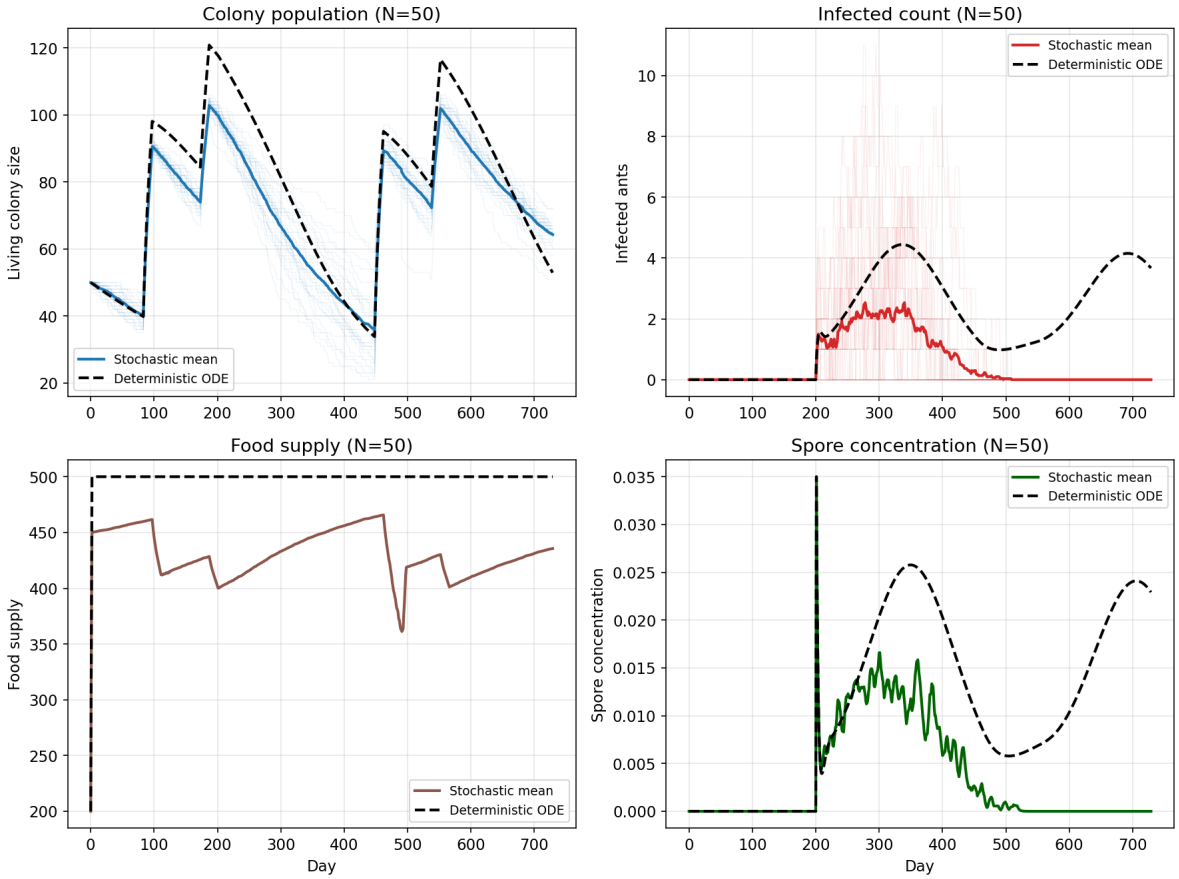

Figure 10: Stochastic vs. deterministic comparison at $N_0 = 50$, $p = 0.15$

At $N_0 = 500$ (Figure 11), the stochastic mean and the deterministic solution nearly overlap during the infection wave. The peak infected count and spore concentration are well matched. This is the law of large numbers, demonstrating how with 500 ants, the fraction infected on any given day is close to $c_n \cdot p$ times the fraction healthy, which is exactly what the ODE computes. However, the discrepancy appears after the crash, when the colony drops to < 10 ants. At that point the ODE allows a fractional amount of ants (e.g., 0.6 ants) to rebuild their population. The stochastic model has the colony either goes extinct or a handful of survivors restart the cycle.

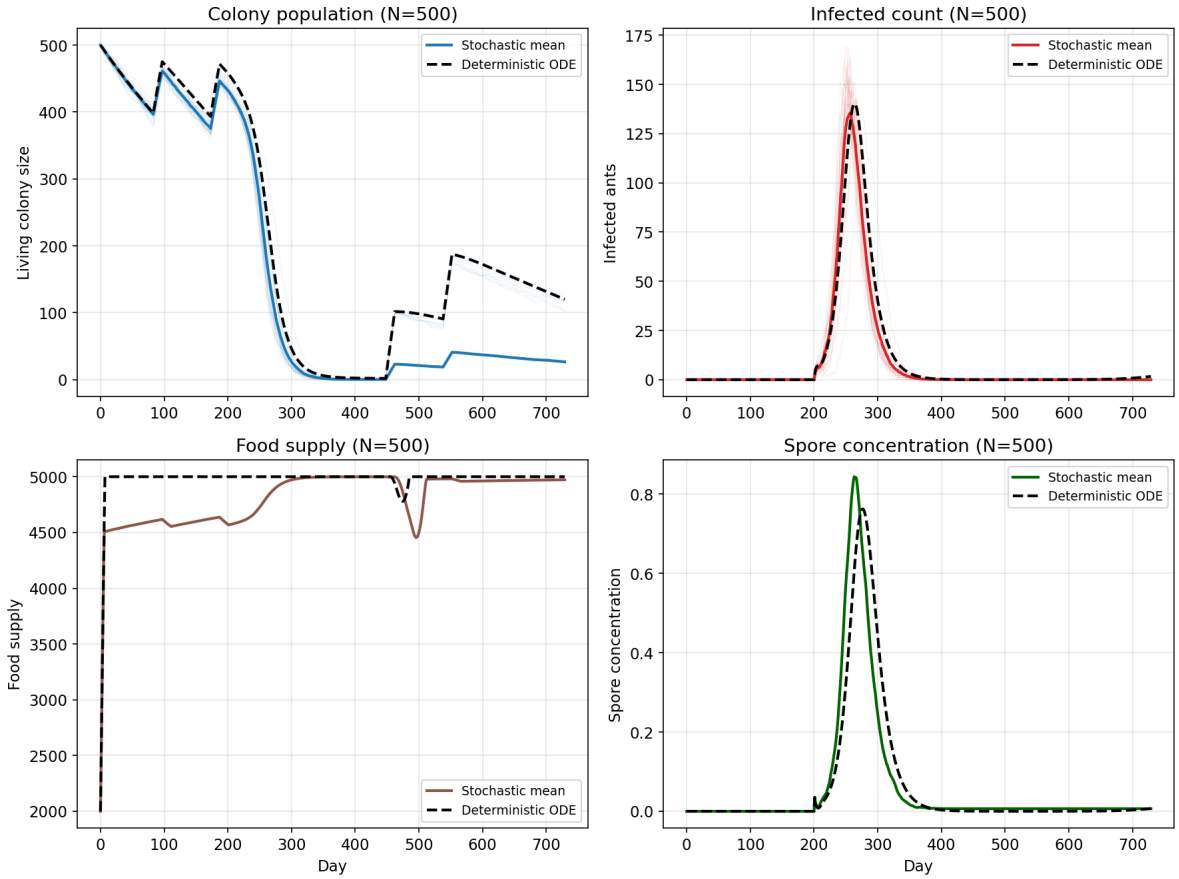

Figure 11: Stochastic vs. deterministic comparison at $N_0 = 500$, $p = 0.15$

7 Conclusions

This project developed an agent based stochastic simulation of Ophiocordyceps unilateralis infecting a carpenter ant colony to determine what controls colony survival during an outbreak. The model shows a sharp threshold near $p \approx 0.15$, which marks the boundary between recovery and collapse. Below this value, sporulation and new infections cannot sustain one another, so the outbreak dies out. Above it, the feedback loop becomes self sustaining and drives the colony toward extinction.

In addition, larger colonies are not more resilient under the $N^{-1/3}$ foraging model because larger infection waves overwhelm the system before recovery can occur. The model also shows that survival is highly sensitive to the daily succumb probability p_s. When p_s increases from 0.01 to 0.04, colony survival shifts from near certainty to near zero, suggesting that improved immune function could greatly improve outcomes.

Finally, comparison with the deterministic ODE model shows agreement at large colony sizes, consistent with the law of large numbers, but divergence at small colony sizes, where discrete ant counts and the outcome of survival or extinction make the stochastic model essential.

References

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